

Instructions. (15 points) Sections 14.6–14.7. Show relevant work for full credit.

(5^{pts}) 1. Let $f(x, y) = x^2y + \sqrt{y}$.

(a) (3 pts) Find the directional derivative of f at the point $(2, 1)$ in the direction toward the point $(5, 5)$.

Solution:

$$\nabla f = \left\langle 2xy, x^2 + \frac{1}{2\sqrt{y}} \right\rangle, \text{ so } \nabla f(2, 1) = \left\langle 4, \frac{9}{2} \right\rangle.$$

$$\text{Direction toward } (5, 5): \mathbf{v} = \langle 3, 4 \rangle, \text{ so } \hat{u} = \frac{1}{5} \langle 3, 4 \rangle.$$

$$D_{\hat{u}}f(2, 1) = \left\langle 4, \frac{9}{2} \right\rangle \cdot \frac{1}{5} \langle 3, 4 \rangle = \frac{1}{5} (12 + 18) = \boxed{6}$$

(b) (2 pts) What is the minimum rate of change of f at $(2, 1)$, and in what direction does it occur?

Solution:

$$\text{Minimum rate of change} = -|\nabla f(2, 1)| = -\sqrt{16 + \frac{81}{4}} = \boxed{-\frac{\sqrt{145}}{2}}.$$

$$\text{It occurs in the direction } -\nabla f(2, 1) = \langle -4, -\frac{9}{2} \rangle \text{ (or the unit vector } \frac{1}{\sqrt{145}} \langle -8, -9 \rangle).$$

(4^{pts}) 2. Find all critical points of $f(x, y) = x^3 - 6xy + 8y^3$ and classify each as a local maximum, local minimum, or saddle point.

Solution:

$$f_x = 3x^2 - 6y = 0 \quad \Rightarrow \quad y = \frac{x^2}{2}$$

$$f_y = -6x + 24y^2 = 0 \quad \Rightarrow \quad x = 4y^2$$

Substituting $x = 4y^2$ into $y = x^2/2$:

$$y = \frac{(4y^2)^2}{2} = 8y^4 \quad \Rightarrow \quad y(8y^3 - 1) = 0$$

So $y = 0$ (giving $x = 0$) or $y = \frac{1}{8}$ (giving $x = 1$).

Critical points: $(0, 0)$ and $(1, \frac{1}{8})$.

$$f_{xx} = 6x, \quad f_{yy} = 48y, \quad f_{xy} = -6.$$

$$D = f_{xx}f_{yy} - (f_{xy})^2 = 288xy - 36.$$

$$\text{At } (0, 0): D = 0 - 36 = -36 < 0 \Rightarrow \boxed{\text{saddle point}}.$$

$$\text{At } (1, \frac{1}{8}): D = 288 \cdot \frac{1}{8} - 36 = 108 > 0 \text{ and } f_{xx} = 6 > 0 \Rightarrow \boxed{\text{local minimum}}.$$

- (6^{pts}) **3.** Find the absolute maximum and absolute minimum values of $f(x, y) = x^2 - 2x + y^2$ on the closed disk $x^2 + y^2 \leq 4$.

Solution:

Interior: $f_x = 2x - 2 = 0 \Rightarrow x = 1$. $f_y = 2y = 0 \Rightarrow y = 0$.

Critical point: $(1, 0)$. Check: $1 + 0 \leq 4$ so it is inside D . $f(1, 0) = 1 - 2 + 0 = -1$.

Boundary: $x = 2 \cos \theta$, $y = 2 \sin \theta$:

$$f = 4 \cos^2 \theta - 4 \cos \theta + 4 \sin^2 \theta = 4 - 4 \cos \theta$$

$$f'(\theta) = 4 \sin \theta = 0 \Rightarrow \theta = 0 \text{ or } \pi$$

At $\theta = 0$: $(2, 0)$, $f = 0$. At $\theta = \pi$: $(-2, 0)$, $f = 8$.

Comparing: $f(1, 0) = -1$, $f(2, 0) = 0$, $f(-2, 0) = 8$.

Absolute max = 8 at $(-2, 0)$. Absolute min = -1 at $(1, 0)$.